SOFTWARE REQUIREMENTS SPECIFICATION

Starlink Wi-Fi Subscription Portal

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for the **Starlink Wi-Fi Subscription Portal**, a captive-portal-based system that allows tenants to purchase time-limited Internet access through a MikroTik router.

The system will allow a tenant to:

1. Connect a phone, laptop, or other supported device to the Wi-Fi network.
2. Be redirected to a captive portal if the device does not have an active subscription.
3. Select one of the available subscription plans.
4. Pay using supported mobile-money services through Fapshi.
5. Have the payment verified by the server.
6. Receive Internet access automatically after successful payment.
7. Use the subscription only from the device associated with that subscription.
8. View the subscription status and expiry time.

The system will also provide an administration portal through which the owner can manage devices, subscriptions, payments, vouchers, and exceptional cases.

The purpose of this document is to provide a single agreed description of the system's behavior so that development, network configuration, testing, deployment, and maintenance can be performed consistently.

1.2 Scope

1.2.1 In Scope

The first release shall include:

- MikroTik captive portal functionality.
- Tenant-facing Wi-Fi subscription website.
- Mobile-first HTML/CSS/JavaScript interface.
- PHP backend without a framework.
- Fapshi mobile-money payment integration.
- MTN Mobile Money and Orange Money support through Fapshi, subject to Fapshi API availability.
- Exactly two subscription plans:
  - 2,000 XAF for 10 days.
  - 5,000 XAF for 30 days.
- Automatic payment verification.
- Automatic MikroTik provisioning after successful payment.
- Single-device subscription binding.
- MAC/device identification.

- Device status and subscription status pages.
- Anti-tethering measures.
- Per-user speed and connection controls where supported by the MikroTik configuration.
- Owner administration portal.
- Device approval.
- Subscription revocation.
- Subscription extension.
- Device re-binding.
- Cash-payment vouchers.
- Payment records.
- Audit logging.
- Payment reconciliation.
- Starlink outage handling.
- Fapshi failure handling.
- Router availability handling.
- Local portal-server deployment.
- Database backups and recovery procedures.

1.2.2 Out of Scope

The first release shall not include:

- Data-volume quotas.
- Usage-based billing.
- Multiple buildings/sites managed from one portal.
- Multiple MikroTik routers managed centrally.
- Native Android/iOS applications.
- Card payments.
- Payment providers other than Fapshi.
- Automatic recurring billing.
- Automatic subscription renewal.
- Self-service extra-device subscriptions.
- Referral systems.
- Advanced revenue analytics.

These features may be considered future enhancements.

1.3 Intended Audience

User	Purpose
Project Owner	Approves requirements and manages the service
Developer	Implements the portal and backend
Network Technician	Configures MikroTik, Starlink and network rules
Tester	Tests the website, payment flow and network behavior
Tenant	Purchases and uses Internet access

1.4 Definitions

Term	Definition
Captive Portal	A webpage shown to a device before Internet access is permitted
Hotspot	MikroTik RouterOS feature used to control authenticated Internet access
Walled Garden	Network destinations allowed before authentication
MAC Address	Network interface identifier used to associate a device with a subscription
Private/Random MAC	A MAC address generated by modern devices for Wi-Fi privacy
Tethering	Sharing a device's Internet connection with another device
TTL	Time To Live value in an IP packet
Fapshi	Payment service used by the system for mobile-money transactions
MoMo	MTN Mobile Money
OM	Orange Money
Voucher	A single-use access code generated by the owner
CGNAT	Carrier-grade NAT used by many Internet providers, including Starlink configurations
Webhook	Server-to-server notification concerning a payment
Subscription	A paid period of Internet access associated with one device
XAF	Central African CFA franc
SRS	Software Requirements Specification

1.5 References

The system shall be developed with reference to:

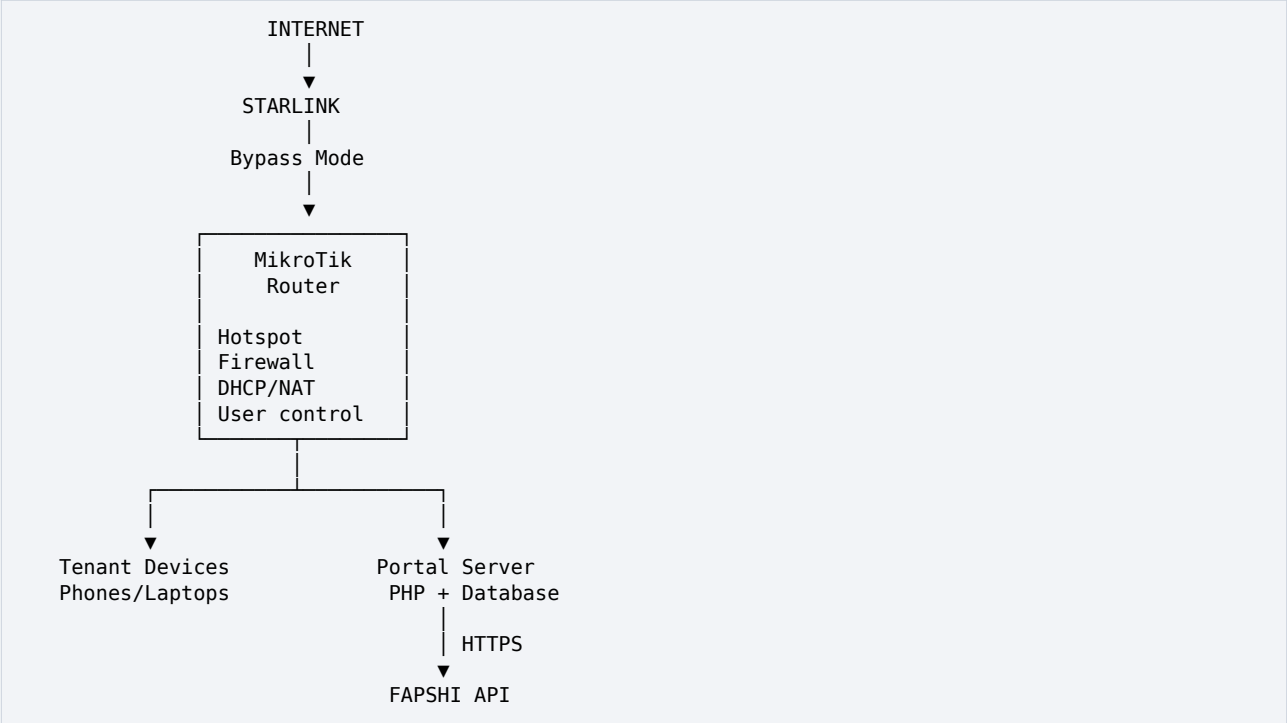
- Fapshi developer documentation.
- MikroTik RouterOS documentation.
- MikroTik Hotspot documentation.
- MikroTik RouterOS API documentation.
- IEEE 830 requirements-specification guidance.
- ISO/IEC/IEEE 29148 requirements-engineering guidance.
- Starlink documentation applicable to the installed dish and router.

Exact Fapshi endpoints, request fields, response fields, authentication mechanism, and payment-status values shall be verified against the current Fapshi documentation before implementation.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The system consists of the following major components:



The MikroTik shall be the device responsible for enforcing Internet access.

The portal server shall handle business logic, payment processing, subscription management, and communication with the MikroTik.

2.2 Core Business Model

The system shall provide exactly two subscription plans.

Plan	Price	Validity
10-Day Plan	2,000 XAF	10 consecutive days
30-Day Plan	5,000 XAF	30 consecutive days

The system shall not display or sell additional plans in the first release.

The owner may later change these values through a future approved requirements change.

2.3 Subscription Clock

The subscription shall begin immediately after successful server-side confirmation of payment.

For example:

If a 10-day subscription is successfully confirmed at:

1 October at 14:30

the subscription shall become active at approximately:

1 October at 14:30

and expire after the defined 10-day period.

The system shall store the start time and expiry time explicitly.

2.4 User Classes

Tenant

A tenant is a person who purchases Internet access.

The tenant interface shall be simple and mobile-first.

Owner/Administrator

The owner manages:

- Devices.
- Subscriptions.
- Payments.
- Vouchers.
- Exceptional access.
- Rebinding.
- Revocation.
- Extension.
- Audit records.

Technician/Developer

The technician or developer is responsible for:

- Initial installation.
- MikroTik configuration.
- Server installation.
- Payment integration.
- Maintenance.
- Troubleshooting.

2.5 Operating Environment

Internet

Starlink shall provide the Internet connection.

Where supported by the installed equipment, the Starlink router shall operate in bypass mode so that the MikroTik becomes the primary gateway for the tenant network.

Gateway

The gateway shall be a MikroTik router running RouterOS 7 or a compatible version capable of providing:

- Hotspot.
- Firewall.
- DHCP.
- NAT.
- User profiles.
- API access.
- Appropriate connection/session controls.

The exact router model shall be confirmed before deployment.

Portal Server

The portal shall run on a local device connected to the MikroTik LAN.

Possible hardware includes:

- Small PC.
- Laptop.
- Mini PC.
- Raspberry Pi or equivalent supported device.

The server may run:

- Linux, or
- Windows.

The software environment shall include:

- PHP 8.x or a compatible supported version.
- Apache or Nginx.
- SQLite or MySQL/MariaDB.

The recommended first-release deployment is **local hosting on the LAN**.

The system shall not require a cloud VPS or external web-hosting subscription for normal operation.

2.6 Cost and Hosting Requirement

The portal shall be designed to operate without recurring web-hosting costs when the required MikroTik, Starlink connection, and local server hardware are already available.

The following shall not be mandatory:

- VPS.
- Cloud hosting.
- Domain name.
- Public IP address.
- External hosting provider.

The portal server shall communicate with Fapshi through outbound HTTPS.

The system shall therefore remain functional behind Starlink CGNAT without requiring incoming Internet connections to the portal server.

Transaction fees charged by the payment provider, if applicable, are outside the software-development cost.

2.7 Assumptions and Dependencies

The system assumes:

- Starlink service is active.
- MikroTik is available and correctly configured.
- The portal server has a stable local IP address.
- The portal server can reach the Internet through the MikroTik.
- The MikroTik clock is synchronized.
- Fapshi's API remains available and compatible.
- Tenants have supported mobile-money accounts.
- Tenant devices can connect to the Wi-Fi network.
- Device MAC addresses remain stable during the subscription unless intentionally changed by the device or user.
- Required power is available or protected by a UPS/battery where possible.

3. SYSTEM ARCHITECTURE

3.1 Components

Component	Responsibility
Starlink	Internet connectivity
MikroTik	Network gateway and access enforcement
Portal Server	Application and business logic
Database	Persistent application data
Tenant Website	Subscription purchase and status interface
Admin Portal	Owner management
Fapshi	Mobile-money payment processing

3.2 Subscription Flow



3.3 Separation of Responsibilities

Website/Backend

The application shall:

- Display plans.
- Create orders.
- Communicate with Fapshi.
- Verify payment.
- Create subscriptions.
- Store records.
- Communicate with MikroTik.
- Provide tenant status.
- Provide administration.
- Record audit events.

MikroTik

The MikroTik shall:

- Provide Wi-Fi/LAN connectivity.
- Redirect unauthenticated devices.
- Enforce access.
- Apply user profiles.
- Enforce expiry.
- Apply speed/connection limits.
- Restrict unauthorized devices.
- Apply firewall and anti-tethering measures.

The website shall not be relied upon as the sole mechanism for preventing Internet access.

4. FUNCTIONAL REQUIREMENTS

4.1 Captive Portal

ID	Requirement	Priority
FR-CP-01	The system shall redirect an unauthenticated device to the captive portal.	Must
FR-CP-02	The system shall prevent an unpaid device from accessing the external Internet.	Must
FR-CP-03	The portal shall allow access to required portal resources before payment.	Must
FR-CP-04	The portal shall display the available subscription plans.	Must
FR-CP-05	The portal shall provide a status page.	Must
FR-CP-06	The portal shall provide a help/troubleshooting page.	Should

4.2 Subscription Plans

ID	Requirement	Priority
FR-PL-01	The system shall provide exactly two plans.	Must
FR-PL-02	The 10-Day Plan shall cost 2,000 XAF.	Must
FR-PL-03	The 10-Day Plan shall provide 10 days of access.	Must
FR-PL-04	The 30-Day Plan shall cost 5,000 XAF.	Must
FR-PL-05	The 30-Day Plan shall provide 30 days of access.	Must
FR-PL-06	The system shall not allow the browser to determine or modify the plan price.	Must
FR-PL-07	The backend shall retrieve the authoritative plan price from the database/configuration.	Must
FR-PL-08	The system shall reject a payment amount that does not match the selected plan.	Must

4.3 Payment

ID	Requirement	Priority
FR-PY-01	The system shall initiate payments through Fapshi.	Must
FR-PY-02	The system shall send the correct amount in XAF.	Must
FR-PY-03	The backend shall associate each payment with a unique order reference.	Must
FR-PY-04	The server shall verify payment status before activating access.	Must
FR-PY-05	The server shall verify that the confirmed amount matches the selected plan.	Must
FR-PY-06	The system shall prevent duplicate payment notifications from creating duplicate subscriptions.	Must
FR-PY-07	The system shall not trust payment-success information supplied only by the browser.	Must
FR-PY-08	The system shall store the Fapshi transaction identifier.	Must
FR-PY-09	Failed payments shall not activate Internet access.	Must
FR-PY-10	Expired or cancelled payments shall not activate Internet access.	Must
FR-PY-11	Fapshi credentials shall never be exposed to tenant browsers.	Must
FR-PY-12	Payment processing errors shall be recorded for troubleshooting.	Must

4.4 Subscription Activation

ID	Requirement	Priority
FR-AP-01	A subscription shall only be created after successful payment confirmation.	Must
FR-AP-02	Each subscription shall have a unique identifier.	Must
FR-AP-03	Each subscription shall record its plan, device, start time and expiry time.	Must
FR-AP-04	The subscription shall begin immediately after payment confirmation.	Must
FR-AP-05	A 2,000 XAF subscription shall expire after 10 days.	Must
FR-AP-06	A 5,000 XAF subscription shall expire after 30 days.	Must
FR-AP-07	The system shall automatically remove/restrict expired access.	Must
FR-AP-08	The system shall not create multiple active subscriptions from the same successful transaction.	Must

4.5 Device Binding

Each paid subscription shall be associated with one authorized device.

ID	Requirement	Priority
FR-DB-01	A subscription shall be bound to one device identifier.	Must
FR-DB-02	A second device shall not use the same subscription.	Must
FR-DB-03	The system shall record the device MAC address when the subscription is activated.	Must
FR-DB-04	The system shall display the associated device to the owner.	Must

ID	Requirement	Priority
FR-DB-05	The owner shall be able to re-bind a subscription to a different device.	Must
FR-DB-06	Re-binding shall be recorded in the audit log.	Must
FR-DB-07	The tenant shall receive a clear message if the device identity has changed.	Should

4.6 Private/Random MAC Handling

Modern phones may use private/randomized MAC addresses.

The system shall therefore:

- Detect the MAC presented by the device to the MikroTik.
- Use the network-specific MAC presented during subscription activation.
- Explain to the tenant that changing/resetting the private Wi-Fi address may cause the device to appear as a different device.
- Provide an owner re-binding mechanism.
- Avoid assuming that the physical hardware MAC is always available to the network.

4.7 Anti-Tethering

ID	Requirement	Priority
FR-AT-01	The system shall implement measures to restrict common Wi-Fi, USB and Bluetooth tethering.	Must
FR-AT-02	The MikroTik shall apply appropriate firewall/connection controls.	Must
FR-AT-03	TTL-based filtering may be used as one anti-tethering mechanism.	Should
FR-AT-04	Anti-tethering shall be tested against supported Android and iOS devices.	Must
FR-AT-05	The system shall document that anti-tethering cannot guarantee protection against every possible sharing technique.	Should

The system shall evaluate anti-tethering based on observed behavior rather than depending exclusively on one technical mechanism.

4.8 Speed and Connection Limits

The MikroTik shall support per-user limitations where technically appropriate.

The exact speed assigned to each plan shall be confirmed before final deployment.

The SRS currently leaves the speed values open because they depend on:

- Available Starlink bandwidth.
- Number of tenants.
- Network conditions.
- Owner's desired service level.

4.9 Tenant Status

The tenant shall be able to view:

- Current plan.
- Payment status.
- Subscription status.
- Start time.
- Expiry time.
- Remaining time.
- Associated device information where appropriate.

The system shall not expose another tenant's information.

4.10 Administration Portal

The owner shall have access to a password-protected administration portal.

The portal shall provide:

Dashboard

- Active subscriptions.
- Today's payments.
- Pending/stuck payments.
- Expired subscriptions.
- Router status.
- Fapshi status.

Devices

The owner shall be able to:

- View connected devices.
- View authorized devices.
- Approve a device.
- Revoke a device.
- Re-bind a subscription.

Subscriptions

The owner shall be able to:

- Search subscriptions.
- View active subscriptions.
- View expired subscriptions.
- Revoke subscriptions.
- Extend subscriptions.
- View associated devices.

Payments

The owner shall be able to:

- View transactions.
- Search by reference.
- Search by phone number where appropriate.
- View payment status.
- View payment amount.
- View transaction date.
- Export payment information where implemented.

Vouchers

The owner shall be able to:

- Generate vouchers.
- View voucher status.
- Redeem/activate cash-payment vouchers.
- Determine whether a voucher has already been used.

Audit Log

The system shall record important administrator actions.

4.11 Cash Vouchers

ID	Requirement	Priority
FR-VC-01	The owner shall be able to generate a voucher.	Must
FR-VC-02	A voucher shall be usable only once.	Must
FR-VC-03	A redeemed voucher shall become permanently unavailable for reuse.	Must
FR-VC-04	Voucher activation shall bind the subscription to one device.	Must
FR-VC-05	Voucher redemption shall be recorded in the audit log.	Must
FR-VC-06	A voucher shall correspond to one of the two defined plans.	Must

4.12 Outage Handling

The system shall handle different types of failures independently.

Starlink outage

If Starlink is unavailable:

- New Internet access shall not be falsely presented as available.
- The portal shall display an appropriate service-status message where possible.
- New payment activation shall be handled safely.
- Payments already made shall be reconciled after connectivity returns.

Fapshi outage

If Fapshi is unavailable:

- New payments shall not be falsely marked successful.
- The tenant shall receive an appropriate message.
- Pending transactions shall be reconciled when service returns.

Portal server outage

Existing MikroTik-side subscriptions should continue to be enforced by the router until their expiry.

The portal server shall not be required for every Internet packet after a subscription has been provisioned.

MikroTik outage

If the MikroTik is unavailable, tenant Internet access shall not be possible through the system.

The application shall record provisioning failures when it can detect them.

4.13 Logging and Reporting

ID	Requirement	Priority
FR-LG-01	The system shall log payment events.	Must
FR-LG-02	The system shall log payment status changes.	Must
FR-LG-03	The system shall log subscription creation.	Must
FR-LG-04	The system shall log administrative actions.	Must
FR-LG-05	The system shall log MikroTik provisioning errors.	Must
FR-LG-06	Secrets and API keys shall not be written to logs.	Must
FR-LG-07	The system should provide payment/subscription reconciliation information.	Should

5. EXTERNAL INTERFACE REQUIREMENTS

5.1 Tenant Interface

The tenant website shall contain:

Home/Plans Page

- 10-Day Plan — 2,000 XAF.
- 30-Day Plan — 5,000 XAF.
- Service-status message.
- Purchase buttons.

Checkout Page

- Selected plan.
- Price.
- Phone number.
- Payment instructions.

- Payment status.
- Retry option.

Status Page

- Active plan.
- Start time.
- Expiry time.
- Remaining time.
- Device status.

Help Page

The page shall explain:

- How to purchase.
- How payment works.
- One-device rule.
- Private MAC considerations.
- What to do if Internet access does not begin.
- How to contact the owner.

5.2 Interface Design Requirements

The tenant interface shall be:

- Mobile-first.
- Responsive.
- Lightweight.
- Easy to understand.
- Usable on low-end Android devices.
- Usable on iPhones.
- Usable on laptops.

The portal shall not depend on:

- Large external images.
- External fonts.
- Third-party JavaScript libraries that are unnecessary for core operation.

This is important because an unpaid device may have access only to the portal server and approved resources.

5.3 Admin Interface

The administration interface shall contain:

- Login.
- Dashboard.
- Devices.
- Subscriptions.
- Payments.
- Vouchers.
- Audit log.
- System/service status.

5.4 Hardware Interfaces

Starlink → MikroTik

Ethernet connection.

MikroTik → Tenant Devices

Wi-Fi and/or Ethernet.

MikroTik → Portal Server

LAN/Ethernet connection.

The portal server shall have a reserved or static local IP address.

5.5 Fapshi Interface

The backend shall communicate with Fapshi using HTTPS.

The exact API implementation shall be isolated inside a dedicated PHP module.

The module shall support the Fapshi operations required for:

- Payment initiation.
- Payment status verification.
- Payment reconciliation.
- Webhook processing if supported/used.

The exact Fapshi API endpoints and fields shall be confirmed from current documentation before implementation.

5.6 MikroTik API Interface

The backend shall communicate with the MikroTik using a secured RouterOS API mechanism.

Required operations may include:

- Creating hotspot users.
- Assigning user profiles.
- Associating users with MAC addresses.
- Removing/revoking users.
- Updating user expiry.

- Reading connected users.
- Checking router status.

Router API credentials shall:

- Be stored outside the public web root.
- Be restricted to the portal server.
- Use the minimum permissions required.
- Never be exposed to tenants.

6. DATA REQUIREMENTS

6.1 Main Entities

The database shall contain, at minimum:

Plans

Fields may include:

- plan_id
- name
- price
- duration_days
- speed profile
- status

The first release shall contain exactly two active plans.

Orders

Fields may include:

- order_id
- order_reference
- plan_id
- device_id
- phone_number
- amount
- status
- Fapshi_transaction_id
- created_at
- updated_at

Subscriptions

Fields may include:

- subscription_id
- order_id
- plan_id
- device_id
- start_time
- expiry_time
- status
- created_at

Devices

Fields may include:

- device_id
- MAC address
- device name/description
- first_seen
- last_seen
- status

Vouchers

Fields may include:

- voucher_id
- code
- plan_id
- status
- redeemed_by_device
- redeemed_at
- created_at

Admin Users

Fields may include:

- admin_id
- username
- password_hash
- status
- created_at

Audit Logs

Fields may include:

- log_id
- administrator
- action
- target
- timestamp
- relevant details

6.2 Database Business Rules

The database shall enforce or the application shall enforce the following:

1. Each successful payment reference shall be unique.
2. A successful payment shall not create multiple subscriptions.
3. A subscription shall have exactly one plan.
4. A subscription shall have one associated device.
5. An active subscription shall have a valid start time and expiry time.
6. An expired subscription shall not remain active.
7. A voucher shall not be redeemable more than once.
8. Payment amounts shall match the plan price.
9. A subscription shall not be activated without successful payment or a valid voucher/owner-approved action.
10. Administrative re-binding shall be logged.

6.3 Data Retention

The proposed initial retention period is **12 months**, after which records may be anonymized or deleted according to the owner's final privacy/retention policy.

The final retention period shall be confirmed before production deployment.

7. MIKROTIK REQUIREMENTS

The MikroTik shall provide:

- DHCP.
- NAT.
- Hotspot.
- Captive portal.
- Walled garden.
- Firewall rules.
- User profiles.
- Session/expiry enforcement.
- Router API.

- Appropriate connection limits.
- Anti-tethering measures.
- Time synchronization.

The router shall be configured so that unpaid devices cannot bypass the captive portal to access external Internet services.

The router shall remain capable of enforcing active subscriptions if the portal server temporarily becomes unavailable.

8. SECURITY REQUIREMENTS

8.1 Authentication

The admin portal shall require authentication.

Passwords shall never be stored in plaintext.

Passwords shall be stored using a secure password-hashing mechanism provided by PHP.

8.2 Payment Security

The system shall:

- Verify payments server-side.
- Validate payment amount.
- Validate transaction identity.
- Prevent replay/duplicate processing.
- Keep API credentials secret.
- Avoid trusting browser-submitted payment-success information.

8.3 Router Security

The router API account shall:

- Be dedicated to the application.
- Have only required permissions.
- Accept API connections only from the portal server where possible.
- Use a strong password.
- Not be exposed to tenant devices.

8.4 Web Security

The application shall protect against:

- SQL injection.
- Cross-site scripting.
- Cross-site request forgery where applicable.
- Session hijacking.

- Unauthorized admin access.
- Parameter manipulation.
- Unauthorized subscription activation.

User input shall be validated and sanitized appropriately.

Database queries shall use prepared statements.

9. NON-FUNCTIONAL REQUIREMENTS

9.1 Performance

ID	Requirement
NFR-PF-01	Normal portal pages should load quickly on a local network.
NFR-PF-02	Payment-status polling shall not unnecessarily overload Fapshi or the local server.
NFR-PF-03	Subscription activation should normally occur within 10 seconds after successful payment confirmation, subject to Fapshi and MikroTik response time.
NFR-PF-04	Expired access should be removed within 2 minutes of expiry.

9.2 Reliability

The system should:

- Recover automatically after server restart.
- Preserve database records after restart.
- Continue router-side enforcement when the portal server is temporarily unavailable.
- Provide backup and restoration procedures.
- Reconcile pending payments after connectivity returns.

9.3 Availability

The local portal server should remain powered while the service is operating.

A UPS or battery backup is recommended for:

- MikroTik.
- Portal server.
- Relevant network equipment.

9.4 Maintainability

The code shall be organized into separate modules for:

- Database.
- Fapshi.
- MikroTik.
- Authentication.

- Subscriptions.
- Orders.
- Administration.

Changes to the Fapshi API shall preferably require modifications only to the Fapshi integration module.

9.5 Usability

The tenant should be able to purchase a subscription without technical knowledge.

The interface shall use:

- Clear buttons.
- Clear prices.
- Simple instructions.
- Visible payment status.
- Clear error messages.

10. PROPOSED TECHNOLOGY STACK

Component	Technology
Frontend	HTML5, CSS3, JavaScript
Backend	PHP 8.x
Backend framework	None
Database	SQLite initially; MySQL/MariaDB if required
Web server	Apache or Nginx
Router	MikroTik
Router OS	RouterOS 7 or compatible
Payment	Fapshi
Communication	HTTPS
Local network	Ethernet/Wi-Fi
Internet	Starlink

The first release shall prioritize simplicity and low operating cost.

11. PROPOSED PROJECT STRUCTURE

```
portal/
│
├── public/
│   ├── index.html
│   ├── checkout.html
│   ├── status.html
│   ├── help.html
│   ├── assets/
│   │   ├── style.css
│   │   └── app.js
│   │
│   └── api/
│       ├── create_order.php
│       ├── payment_status.php
│       ├── fapshi_webhook.php
│       └── subscription_status.php
│
├── admin/
│   ├── login.php
│   ├── dashboard.php
│   ├── devices.php
│   ├── subscriptions.php
│   ├── payments.php
│   ├── vouchers.php
│   └── audit.php
│
├── src/
│   ├── config.php
│   ├── db.php
│   ├── auth.php
│   ├── fapshi.php
│   ├── mikrotik.php
│   ├── subscriptions.php
│   ├── payments.php
│   └── audit.php
│
├── cron/
│   ├── reconcile.php
│   └── expire.php
│
├── storage/
│   └── database.sqlite
│
└── logs/
```

Sensitive configuration and database files shall remain outside the publicly accessible web root where possible.

12. USE CASES

UC-01: Purchase Subscription

Actor: Tenant

Preconditions

- Tenant is connected to the Wi-Fi.
- Starlink and MikroTik are operational.
- Tenant does not have active access.

Main Flow

1. Tenant connects to Wi-Fi.
2. MikroTik redirects tenant to the captive portal.

3. Tenant views the two plans.
4. Tenant selects a plan.
5. Tenant enters the required mobile-money information.
6. Backend creates an order.
7. Backend initiates payment through Fapshi.
8. Tenant approves the payment.
9. Backend verifies successful payment.
10. Backend creates the subscription.
11. Backend associates the subscription with the tenant's device.
12. Backend provisions the MikroTik.
13. Tenant receives Internet access.
14. Tenant can view subscription status.

Alternative Flow

If payment fails:

- No subscription is created.
- No Internet access is granted.
- The tenant receives an appropriate failure message.

UC-02: Subscription Expiry

1. Subscription reaches expiry time.
2. Backend/router identifies the subscription as expired.
3. Internet access is removed.
4. Device is redirected to the portal.
5. Tenant can purchase another subscription.

UC-03: Owner Rebinds Device

1. Owner logs into admin portal.
2. Owner searches for subscription.
3. Owner selects re-bind.
4. Owner selects/provides the new device.
5. System removes authorization from the previous device.
6. System associates the subscription with the new device.
7. System provisions MikroTik.
8. System records the action in the audit log.

UC-04: Owner Issues Voucher

1. Owner logs into admin portal.
2. Owner selects voucher generation.
3. Owner selects one of the two plans.
4. System generates a unique voucher.
5. Owner receives the voucher code.
6. Tenant enters the voucher.
7. System verifies that the voucher is unused.
8. Subscription is created.
9. Voucher is marked used.
10. Subscription is bound to the tenant's device.

UC-05: Payment Reconciliation

1. System identifies pending payment.
2. System queries Fapshi.
3. System receives current transaction status.
4. If payment is successful, the system verifies the amount.
5. System creates the subscription if one does not already exist.
6. System provisions MikroTik.
7. System records the reconciliation event.

13. RISK ANALYSIS

Risk	Impact	Mitigation
Tethering	Revenue loss/congestion	Anti-tethering controls and testing
Private MAC changes	Device loses access	Clear tenant instructions and admin rebind
Power outage	Service interruption	UPS/battery backup
Starlink outage	No Internet	Outage handling and reconciliation
Fapshi outage	Payment unavailable	Pending status, reconciliation, vouchers
Duplicate payment notifications	Duplicate access	Idempotency and transaction uniqueness
Router API credential leak	Network compromise	Restricted account and protected configuration
MAC cloning	Unauthorized access	Device binding and session controls
Captive portal detection failure	Tenant confusion	Manual portal address/help page
VPN/private DNS	Portal problems	Troubleshooting instructions
Portal server failure	New purchases unavailable	Backups and router-side enforcement

14. TESTING REQUIREMENTS

14.1 Functional Tests

The following shall be tested:

- Unpaid device cannot browse the Internet.
- Portal loads.
- Both plans are displayed correctly.
- 2,000 XAF payment activates 10-day plan.
- 5,000 XAF payment activates 30-day plan.
- Incorrect payment amount is rejected.
- Failed payment does not activate access.
- Duplicate notification does not create another subscription.
- Subscription expires correctly.
- Second device cannot use the subscription.
- Rebinding works.
- Voucher works once.
- Voucher cannot be reused.
- Admin revocation works.
- Admin extension works.

14.2 Device Tests

Testing shall include at least:

- Two Android devices from different manufacturers.
- iPhone.
- Windows laptop.
- macOS laptop where available.

Testing shall include devices using private/randomized Wi-Fi MAC addresses.

14.3 Security Tests

The system shall be tested for:

- SQL injection.
- XSS.
- Unauthorized admin access.
- Parameter manipulation.
- Payment replay.
- Forged payment notifications.
- Access before payment.

- Second-device access.
- Router API exposure.
- Credential exposure.

14.4 Failure Tests

The system shall be tested with:

- Starlink disconnected.
- MikroTik unavailable.
- Portal server switched off.
- Fapshi unavailable.
- Payment declined.
- Payment pending.
- Repeated payment notification.
- Server restart.
- Router restart.
- Power interruption.

15. ACCEPTANCE CRITERIA

ID	Acceptance Criterion
AC-01	An unpaid device cannot access external websites/apps and can access only the permitted portal resources.
AC-02	After successful payment confirmation, the appropriate subscription is provisioned and Internet access becomes available normally within 10 seconds, subject to external service/network response time.
AC-03	A 2,000 XAF successful payment creates a 10-day subscription and no other plan.
AC-04	A 5,000 XAF successful payment creates a 30-day subscription and no other plan.
AC-05	A second device cannot use a subscription purchased and bound to the first device.
AC-06	Standard tethering tests using Wi-Fi and USB tethering are blocked or restricted according to the defined anti-tethering test cases.
AC-07	A repeated or forged payment notification creates no duplicate subscription and grants no unauthorized access.
AC-08	An expired subscription loses Internet access within 2 minutes of expiry.
AC-09	The owner can re-bind an active subscription to another device and the previous device loses authorization.
AC-10	The owner can approve an authorized laptop/device in under one minute.
AC-11	The owner can revoke an active device immediately.
AC-12	With the portal server switched off, an already-provisioned subscription remains enforced by the MikroTik until expiry, subject to router operation.

ID	Acceptance Criterion
AC-13	With Starlink unavailable, the system does not falsely report Internet availability and handles payment reconciliation safely.
AC-14	Fapshi credentials do not appear in browser source, client-side JavaScript, responses, or normal application logs.
AC-15	A tenant cannot access the administration interface.
AC-16	A cash voucher works once, creates the appropriate subscription, and cannot be reused.
AC-17	Database records survive a normal portal-server restart.
AC-18	The system can restore the database from a valid backup.

16. DEPLOYMENT

16.1 Initial Deployment

The deployment shall consist of:

1. Install and test Starlink.
2. Configure Starlink bypass mode where applicable.
3. Connect Starlink to MikroTik WAN.
4. Configure MikroTik LAN.
5. Configure DHCP.
6. Configure NAT.
7. Configure Hotspot.
8. Configure the captive portal.
9. Configure walled-garden access.
10. Configure RouterOS API.
11. Configure firewall and anti-tethering controls.
12. Install local portal server.
13. Install PHP.
14. Install Apache/Nginx.
15. Install database.
16. Deploy application.
17. Configure Fapshi credentials.
18. Configure MikroTik API credentials.
19. Create database schema.
20. Create the two subscription plans.
21. Test payment flow.
22. Test device binding.
23. Test subscription expiry.

24. Test administrator functions.
25. Test failure scenarios.
26. Configure backups.
27. Train the owner.

16.2 Backup

The system shall support:

- Regular database backups.
- Router configuration backups.
- Recovery instructions.

The owner should maintain at least one backup separate from the portal server.

16.3 Maintenance

Task	Frequency
Check failed/stuck payments	Daily
Payment reconciliation	Daily automatic
Manual reconciliation review	Weekly
Database backup	Daily
Router configuration backup	After major changes
Audit-log review	Weekly
Service/network review	Monthly
Software/security updates	Planned maintenance
Backup restoration test	Every three months

17. FUTURE ENHANCEMENTS

The following are intentionally excluded from Version 1.1 but may be added later:

- Extra-device subscription.
- SMS reminders.
- WhatsApp notifications.
- Data-volume plans.
- Usage analytics.
- Revenue reports.
- Multiple MikroTik routers.
- Multiple buildings.
- Referral discounts.
- Promotional codes.

- Automatic renewal.
- Customer accounts.
- Native mobile application.

18. TRACEABILITY

Business Objective	Requirements
Tenants can purchase Wi-Fi automatically	FR-CP, FR-PL, FR-PY, FR-AP
Two simple subscription options	FR-PL-01 to FR-PL-08
Access lasts for purchased period	FR-AP-04 to FR-AP-07
One payment covers one device	FR-DB, FR-AT
Owner can handle exceptional cases	FR-AD, FR-VC
Payments cannot be falsely duplicated	FR-PY-04 to FR-PY-12
Service handles outages safely	FR-OU
Tenant information is protected	NFR-SC
System has low operating cost	Section 2.6
System can operate without cloud hosting	Section 2.6
System remains maintainable	Section 9.4

19. FINAL CONFIGURATION DECISIONS

The following decisions are now fixed for this version:

Item	Decision
Number of plans	Exactly 2
10-day plan	2,000 XAF
30-day plan	5,000 XAF
Subscription start	Immediately after confirmed payment
Payment provider	Fapshi
Gateway	MikroTik
Internet source	Starlink
Backend	Raw PHP
Frontend	HTML/CSS/JavaScript

Item	Decision
Hosting	Local server
Cloud hosting	Not required
Domain	Not required
Card payment	Not included
Recurring billing	Not included
Multi-site support	Not included
Data-volume billing	Not included
Device model	One authorized device per subscription

20. ITEMS TO CONFIRM BEFORE PRODUCTION

The main requirements are now defined, but the following technical details must still be confirmed before implementation:

OD-01 — MikroTik model

The exact MikroTik model must be identified.

OD-02 — RouterOS version

The exact RouterOS version must be confirmed.

OD-03 — Starlink hardware

The exact Starlink dish/router generation and Ethernet arrangement must be confirmed.

OD-04 — Plan speeds

The owner must determine the intended speed profile for:

- 10-Day Plan.
- 30-Day Plan.

The prices and durations are already fixed.

OD-05 — Fapshi API implementation

The developer must confirm the current:

- API endpoints.
- Authentication method.
- Payment initiation method.
- Payment status method.
- Webhook behavior.
- Supported mobile-money networks.
- Error/status values.

OD-06 — Interface language

The initial release should be confirmed as either:

- English only, or
- English and French.

OD-07 — Data retention

The final payment/device-data retention period must be approved.

OD-08 — Portal server hardware

The actual local server device must be selected and tested.

21. APPROVAL

This SRS becomes the baseline for development once the remaining technical decisions have been confirmed.

Role	Name	Signature	Date
Project Owner	Mbuh Favour	_____	_____
Developer	_____	_____	_____
Network Technician	_____	_____	_____
Tester	_____	_____	_____

— END OF SOFTWARE REQUIREMENTS SPECIFICATION —

Starlink Wi-Fi Subscription Portal — Version 1.1